

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination Summer 2024 Course: B. Tech. Branch : Computer Engg. and Allied branches Semester : III Subject Code & Name: Discrete Mathematics BTCOC302/ BTCOC302_Y18 Max Marks: 60 Date: 02/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Consider the sets $U = \{1,2,3,...,8,9\}$ and $A = \{1,2,5,6\}$, $B = \{2,5,7\}$, $C = \{1,3,5,7,9\}$ i) Find a) $A \cap B$ and $A \cap C$ b) $A \cup B$ and $B \cup C$	Apply	6
B)	Prove by mathematical induction for all positive integers. $\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$	Apply	6
C)	Prove that $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent using truth table.	Apply	6
Q.2	Solve Any Two of the following.		12
A)	Determine whether the following functions are one-one, onto or both. Justify your answer i) $X = Y = \{a,b,c,d\}$ and $A = \{(a,a),(b,c),(c,d),(d,b)\}$ ii) $X = \{1,2,3\}$ and $Y = \{a,b,c,d\}$ $B = \{(1,a),(2,a),(3,c)\}$	Apply	6
B)	Let set $A = \{1, 2, 3\}$, $B = \{a, b, c\}$ & $C = \{x,y,z\}$. Consider the following relations R & S from A to B and B to C respectively. $R = \{(1,b), (2,a), (2,c)\}$ & $S = \{(a,y), (b,x), (c,y), (c,z)\}$ (i) Find composition relation $R \circ S$. (ii) Write matrices MR, MS & $MR \circ S$ of relations R, S & $R \circ S$.	Apply	6
C)	Solve $a_r = a_{r-1} - 6a_{r-2} = -30$ given $a_0 = 20$, and $a_1 = 5$.	Understand	6
Q. 3	Solve Any Two of the following.		12
A)	Define the following terms i) Planar Graph ii) Bipartite graph iii) Complete Graph	Remember	6
B)	Find the sum of in-degree and Out degree of each vertex from the given graph	Apply	6

C)	Let G be a planar graph with 10 vertices, 3 components and 9 edges. Find the number of regions in G .	Apply	6
Q.4	Solve Any Two of the following.		12
A)	Construct the minimum spanning tree (MST) for the given graph using Prim's Algorithm.	Apply	6
B)	Explain Binary search tree with an example in detail.	Remember	6
C)	Prove that following graph is Hamiltonian or not by Ore's Theorem.	Apply	6
Q. 5	Solve Any Two of the following.		12
A)	Define the following terminology i) Monoid ii) Abelian group iii) Ring	Remember	6
B)	Explain the Disjunctive and Conjunctive Normal Form	Remember	6
C)	Write a short note on Algebraic Structures with two Binary Operation.	Remember	6
	*** End ***		